UMBC Anagrammer

- A command-line C++ anagram game developed for CMSC 202 at UMBC in UMBC Linux Server
- **Project Structure**
    - ├── proj1.cpp
    - ├── proj1_data.txt
    - ├── proj1_sample.txt
- **Background**
    - An anagram is a word or phrase formed by rearranging the letters of another word or phrase, typically using all the original letters exactly once. A partial anagram is when you don't use all the original letters exactly once. Anagramming words is a linguistic puzzle that involves rearranging the letters of a word or phrase to create entirely new combinations of characters, while still using all the original letters. This wordplay not only tests one's vocabulary and cognitive agility but also tests patience within the jumble of letters. For this project, we are going to be creating a game where the computer will choose a word with more than 8 letters. The computer will build a list of all words that can be built from the chosen word (considered a partial anagram). The players will then guess a word. The program will check three things: 1. If the guess is a word from the list, 2. If the guess can be made from the chosen word, and 3. If the guess has not already been guessed. If all three are true, it increments the number of correct anagrams. If any of the three tests fails, it tells the user why and asks for another word. The game ends when all of the partial anagrams have been guessed.
- **Assignment Description**
    - Your assignment is to develop the code to read in a list of words into an array of strings. The program will randomly choose a word that is greater than or equal to 8 characters. The program will then build a list of all words that can be made from the chosen word. The user will guess words which will either be valid partial anagrams or not. Words can only be guessed once otherwise the player will be notified that the word has already been guessed. The game ends when all possible partial anagrams have been guessed.
- **Sample Input and Output**
    - **To compile the program in Linux Type:**
        - g++ -Wall proj1.cpp -o proj1
    - **To run the command in Linux Type:**
        - ./proj1
    - In the screenshot below, the user input in highlighted below for clarity

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[kpatel38@linux3 ~/Documents]$ ./proj1
Welcome to UMBC Anagrammer!
Starting word: religions
What word would you like to try?
religions
religions is an angram of religions
You have guessed 1 of 93 anagrams available
What word would you like to try?
region
region is an angram of religions
You have guessed 2 of 93 anagrams available
What word would you like to try?
gion
gion is not a valid word
You have guessed 3 of 93 anagrams available
What word would you like to try?
legion
legion is not a valid word
You have guessed 3 of 93 anagrams available
What word would you like to try?
ion
ion is an angram of religions
You have guessed 3 of 93 anagrams available
What word would you like to try?
li
Minimum of 3 characters.
What word would you like to try?
region
Already Guessed
You have guessed 4 of 93 anagrams available
What word would you like to try?
rely
rely is an angram of religions
You have guessed 4 of 93 anagrams available
What word would you like to try?
lig
lig is not a valid word
You have guessed 4 of 93 anagrams available
What word would you like to try?
religion
religion is an angram of religions
You have guessed 4 of 93 anagrams available
What word would you like to try?
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